

Comparing machine and human reviewers to evaluate the risk of bias in randomized controlled trials

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Background: Evidence from new health technologies is growing, along with demands for evidence to inform policy decisions, creating challenges in completing health technology assessments (HTAs)/systematic reviews (SRs) in a timely manner. Software can decrease the time and burden by automating the process, but evidence validating such software is limited. We tested the accuracy of RobotReviewer, a semi-autonomous risk of bias (RoB) assessment tool, and its agreement with human reviewers.

Methods: Two reviewers independently conducted RoB assessments on a sample of randomized controlled trials (RCTs), and their consensus ratings were compared with those generated by RobotReviewer. Agreement with the human reviewers was assessed using percent agreement and weighted kappa (κ). The accuracy of RobotReviewer was also assessed by calculating the sensitivity, specificity, and area under the curve in comparison to the consensus agreement of the human reviewers.

Results: The study included 372 RCTs. Inter-rater reliability ranged from $\kappa = -0.06$ (no agreement) for blinding of participants and personnel to $\kappa = 0.62$ (good agreement) for random sequence generation (excluding overall RoB). RobotReviewer was found to use a high percentage of “irrelevant supporting quotations” to complement RoB assessments for blinding of participants and personnel (72.6%), blinding of outcome assessment (70.4%), and allocation concealment (54.3%).

Conclusion: RobotReviewer can help with risk of bias assessment of RCTs but cannot replace human evaluations. Thus, reviewers should check and validate RoB assessments from RobotReviewer by consulting the original article when not relevant supporting quotations are provided by RobotReviewer. This consultation is in line with the recommendation provided by the developers.

KEYWORDS

artificial intelligence, health technology assessment (HTA), inter-rater reliability, randomized controlled trial, risk of bias, systematic review

1 | INTRODUCTION

Knowledge synthesis products such as health technology assessments (HTAs) and systematic reviews (SRs) are

important sources of high-quality information for policy makers, and the production of these assessments and reports requires expertise, human resources, and time.¹ Given the proliferation of health technologies and

published primary research on them, the need for HTAs/SRs will likely increase in the future,² exacerbating existing challenges of synthesizing the evidence in a timely manner.

In recent years, researchers have explored the application of computing and informatics technologies to decrease the time and resource burden of SRs.^{3–8} Despite rapid advancement in this area, an artificial intelligence tool able to perform all stages of the HTA/SR process has yet to be developed. In addition, most tools remain in the pilot phase, with limited evidence of their reliability and validity.

RobotReviewer is a machine-learning tool that assesses the risk of bias (RoB) of randomized clinical trials. It has gained the attention of knowledge synthesis groups such as the Cochrane Collaboration⁹ and research evidence services such as the TRIP Medical Database.¹⁰ To our knowledge, RobotReviewer's accuracy has only been tested by the developers¹¹ and one independent research team. This team¹² concluded that the agreement between the tool and human reviewers is comparable to that between research groups.¹³ However, this study used dichotomous scoring (low vs high/unclear RoB), rather than the three categories (low, unclear, and high RoB) used by the Cochrane Collaboration and others, and thus has limited applicability for those following Cochrane methods. In addition, the study did not explore the relevance of the key sentences provided by RobotReviewer or their utility in quickly confirming RobotReviewer's RoB assessments. RobotReviewer provides three quotations alongside each RoB domain. The sentences are judged by RobotReviewer as the three most informative sentences for each domain which in some cases provides rationale for the RoB assessment for the domain. If none of the supporting quotations are relevant, the RoB assessments become more difficult to validate as no clear rationale for its classification is provided.

Our study expands on the previous study¹² by examining agreement between human reviewers and RobotReviewer using both direct RoB assessments from RobotReviewer (low vs high/unclear RoB) and those converted to meet SR requirements (low, unclear, or high RoB). In addition, we evaluated the relevance of the quotations provided by RobotReviewer judged most informative in RoB decisions. Our objectives were to determine the following aspects:

1. the accuracy and reliability of RoB assessments from RobotReviewer compared to human reviewers, using three (low, unclear, high) and two (low, high/unclear) separate categories of assessment;
2. the ability of RobotReviewer to correctly identify low or high RoB; and
3. the relevance of the quotations used by RobotReviewer in decisions.

What is already known?

- RobotReviewer is a machine learning system for assessing the RoB of randomized clinical trial reports and has attracted the attention of several knowledge synthesis groups. Information has been reported that the agreement between the tool and human reviewers is comparable to that between research groups. However, that particular evaluation was performed only in a previous study published in 2017.

What is new?

- Although RobotReviewer has been reported to have comparable agreement to that between research groups, our analysis found that the supposedly relevant quotations used by RobotReviewer were irrelevant in most of the cases for some domains of the risk of bias tool. Therefore, the use of RobotReviewer in isolation as a second reviewer is not recommended at this point. This assessment is in line with developers' recommendations.

Potential impact?

- The results of this study will motivate research teams working on artificial intelligence in this area to improve and refine products/strategies used that enable the timely completion of HTAs/SRs to meet the increased demand of synthesized products in health care.
- Most of the methods using artificial intelligence applied to health technology assessments (HTA) and systematic reviews are in early stages of investigation, and more empirical evidence is needed to determine their usability in practice.

2 | METHODS

2.1 | Study selection

We analyzed a sample of physical therapy trials that were included in systematic reviews published in the Cochrane Database of Systematic Reviews (CDSR). The full-text of each RCT was obtained. Methods for the identification and selection of studies have been described previously.^{13–16}

2.2 | RoB assessment: Human reviewers

We evaluated the RoB of each RCT using the six-domain (seven-item) RoB tool proposed by the Cochrane Collaboration.¹⁷ We followed the Cochrane Collaboration guidelines to perform RoB assessments and developed decision rules as described in Armijo-Olivo et al.¹³ The following overall ratings were used:

1. low RoB: if all domains were rated as low;
2. unclear RoB: if at least one domain was rated as unclear, and the others as unclear or low; or
3. high RoB: if at least one domain was rated as high.^{18,19}

2.2.1 | Characteristics of the human reviewers

Six reviewers completed RoB assessments in this study. Two held a Bachelor's degree in Health Sciences, one had a Master's in Public Health, one was a PhD student in Orthodontics at the time of the study, and two were trained physical therapists, with doctorates in rehabilitation sciences and 10 years of experience in critical appraisal and systematic reviews. Four of the reviewers had formal training in critical appraisal and systematic reviews. The other two had at least one year of hands-on experience conducting systematic reviews. Two independent, trained reviewers assessed the RoB of all RCTs. The final scoring for each domain was based on consensus between the two reviewers.

2.2.2 | Training process

All reviewers were trained and received guidelines for RoB assessments from the first author (SAO). Reviewer training was carried out using 10 trials not included in the study. Results of the RoB assessments for these 10 studies were independently reviewed and discussed in a group meeting to determine consistency in ratings. In addition, the team members met on a regular basis to further calibrate RoB assessments throughout the study.

2.3 | RoB assessment: RobotReviewer

We used RobotReviewer (<https://robot-reviewer.vortextextsystems/>), a free, online tool, that uses machine-learning to automatically evaluate the RoB of RCTs. The tool was developed based on a training set of over over 12 000 trials (published in pdf) from the Cochrane Database of Systematic Reviews (CDSR).¹¹

RobotReviewer performs two tasks: (a) determines the risk of bias of each domain (low vs high/unclear) and (b) identifies informative text from the manuscript that supports the judgments (if available); RobotReviewer retrieves three candidate sentences per each domain. These sentences can be used by human reviewers to quickly identify informative sentences that may support RobotReviewer's assessment.

Data were collected between December 2017 and January 2018. The tool requires users to upload a portable document format (PDF) file for each RCT into its interface. RobotReviewer then analyzes the file(s) and creates output, including RoB assessments and three quotations used for each domain to support those assessments. This output was transferred to a Microsoft Excel worksheet. Using the abstracted quotations and the Cochrane Collaboration guidelines, we evaluated any domains judged by RobotReviewer as high/unclear RoB and reclassified them as high or unclear RoB following the Cochrane guidelines. In addition, by using the output from RobotReviewer, we classified the total risk of bias of each article according to our team's decision rules as described above.¹³ It is important to note that RobotReviewer was not designed with this method of assessing "overall" bias, and this methodology was specifically generated by the authors of this study to allow comparison between the methods.

If *none of the three quotations* provided by RobotReviewer was found to be relevant to the domain, the RoB was marked as unclear, and the quotation was marked as irrelevant. For studies that RobotReviewer could not initially read (eg, photocopies), we used the Adobe Pro text recognition tool to facilitate RobotReviewer's information extraction. In addition, we completed a subgroup analysis to determine the reliability of RobotReviewer when using PDF files compared to documents converted to text files with optical character recognition software before being reconverted to PDF files (see online Appendix S1 for methods and results).

2.4 | Statistical analysis

2.4.1 | Agreement between RobotReviewer and human reviewers

We determined the agreement between RobotReviewer and the consensus of a panel of human reviewers²⁰⁻²² using percent agreement and Cohen's weighted kappa coefficient (κ) for ordinal data. Analyses were conducted using three categories (low, unclear, high) following common standards for risk of bias assessments and also two categories (low, high/unclear) to evaluate the way that RobotReviewer

provides the assessments. Inter-rater scores for individual domains and overall RoB were considered.

We used the criteria proposed by Byrt²³ to interpret the kappa values.

2.4.2 | Accuracy of RobotReviewer

In order to evaluate the accuracy of the RobotReviewer, we dichotomized all RoB ratings to cross-tabulate the results from the human reviewers and RobotReviewer. We then calculated the sensitivity, specificity, positive and negative predictive values (PPV and NPV), and positive and negative likelihoods ratios (LR+ and LR−) of the RobotReviewer assessments for each domain following established standards for diagnostic accuracy.²⁴ Specifically, we analyzed the ability of RobotReviewer to correctly assess RoB in two ways, low vs high/unclear RoB and high vs low/unclear RoB. True positives or negatives occurred when the rating provided by RobotReviewer agreed with that of the human reviewers. False positives and negatives occurred when RobotReviewer was discordant with the human reviewers.

2.4.3 | Relevance of abstracted quotations

For each domain, we calculated the percentage of irrelevant quotations identified by RobotReviewer. As explained above, for each domain, if *none of the three quotations* provided by RobotReviewer was found to be relevant to the domain, the study was marked to have irrelevant quotations for that domain. The percentage of studies where RobotReviewer provided irrelevant quotations but classified RoB as low was also computed by each domain.

Analyses were conducted using Stata (version 12, StataCorp LLC, College Station, Texas) and Microsoft

Office Excel (version 2016, Microsoft Corporation, Redmond, Washington).

3 | RESULTS

3.1 | Characteristics of included studies

From 43 meta-analyses, 393 RCTs were identified. They assessed the impact of physical therapy interventions for various conditions (eg, musculoskeletal, cardiorespiratory) on outcomes of interest (eg, activity, acute length of stay in hospital). Of the 393 RCTs, 372 were included. RobotReviewer was not able to analyze the files of those excluded. Of the 372 RCTs analyzed, 339 were original PDF files and 33 (9%) were converted using Adobe Pro.

3.2 | Agreement between RobotReviewer and human reviewers

3.2.1 | Two categories (low, high/unclear)

Percent agreement

Using two categories (low RoB or high/unclear RoB), we calculated the percent agreement between RobotReviewer and human reviewers. Agreement ranged from 56.45% (for blinding of participants and personnel) to 97.04% (for overall RoB; see Table 1 for details of agreement in each one of the domains of the RoB tool).

Inter-rater reliability

Weighted kappa was calculated to assess inter-rater reliability between RobotReviewer and the human reviewers and ranged from $\kappa = -0.06$ (no agreement) for blinding of participants and personnel to $\kappa = 0.62$ (good agreement) for random sequence generation (excluding overall RoB) (see Table 1).

TABLE 1 Percent agreement and weighted kappa between RobotReviewer and human reviewers

RoB domain	Two domains (low:high/unclear)		Three domains (low:unclear:high)	
	% Agreement (unweighted)	Kappa (95% CI)	% Agreement (unweighted)	Kappa (95% CI)
Random sequence generation	81.45%	0.6158 (0.534, 0.697)	79.84%	0.6082 (0.561, 0.636)
Allocation concealment	79.84%	0.4084 (0.305, 0.512)	78.49%	0.4017 (0.391, 0.483)
Blinding of participants and personnel	56.45%	0.0594 (0.002, 0.117)	46.24%	0.1207 (0.078, 0.158)
Blinding of outcome assessment	68.55%	0.2319 (0.126, 0.337)	59.95%	0.1965 (0.172, 0.241)
Overall RoB	97.04%	−0.0139 (−0.029, 0.001)	70.43%	0.1508 (0.129, 0.186)

Abbreviation: RoB, risk of bias.

3.2.2 | Three categories (low, unclear, high)

Percent agreement

Overall, percent agreement was lower for each domain after reclassifying high/unclear assessments provided by RobotReviewer. Percent agreement ranged from 46.24% (for blinding of participants and personnel) to 79.84% (for random sequence generation; see Table 1).

Inter-rater reliability

Inter-rater reliability as indicated by weighted kappa was similar, better, or a bit worse after high/unclear judgments were reclassified. Weighted kappa ranged from $\kappa = 0.12$ or poor agreement (for blinding of participants and personnel) to $\kappa = 0.61$ or good agreement (for random sequence generation; see Table 1).

3.3 | Sensitivity and specificity of RobotReviewer

3.3.1 | Low vs high/unclear

Table 2 shows the sensitivity and specificity values of RobotReviewer for discriminating between low vs high/unclear RoB. The parameters of accuracy indicate that RobotReviewer was reasonable at discriminating between low vs high/unclear risk of bias for random sequence generation and allocation concealment domains, but performed poorly for blinding of participants and personnel, blinding of outcome assessment, and overall risk of bias. For all results from this analysis, see Table 2.

TABLE 2 Sensitivity and specificity of RobotReviewer at detecting low RoB

RoB domain	Sensitivity (95% CI)	Specificity (95% CI)
Random sequence generation	0.775 (0.7, 0.839)	0.842 (0.787, 0.887)
Allocation concealment	0.800 (0.663, 0.9)	0.798 (0.75, 0.841)
Blinding of participants and personnel	0.100 (0.0593, 0.155)	0.955 (0.917, 0.979)
Blinding of outcome assessment	0.485 (0.383, 0.587)	0.758 (0.703, 0.808)
Overall RoB	0.000 (0, 0.602)	0.981 (0.961, 0.992)

Abbreviation: RoB, risk of bias.

3.3.2 | High vs low/unclear

Following reclassification of high/unclear RoB assessments into high and low/unclear, we calculated the ability to correctly identify high RoB using only RobotReviewer output. Table 3 shows the sensitivity and specificity values of RobotReviewer for discriminating between high vs low/unclear RoB. Taken together, the parameters indicate that RobotReviewer performed reasonably at discriminating between high vs low/unclear risk of bias for random sequence generation but performed poorly for all other domains.

3.4 | Relevance of abstracted quotations

Between 17.5% and 72.6% of the abstracted quotations (three quotations) were considered irrelevant to the domain. (17.5% for random sequence generation, 72.6% for blinding of participants and personnel, 54.3% for allocation concealment, and 70.4% for blinding of outcome assessment). This finding means that none of the three quotations provided by RobotReviewer was able to help support the risk of bias assessment for that specific domain. Below are examples of the irrelevant supporting quotations provided by RobotReviewer for each one of the domains. It is important to note that none of the other two quotations provided by RobotReviewer in these domains was relevant to make a decision for the specific domain:

- *Random sequence generation*: “Seventy-one patients were entered into the study, with 38 in the intervention group and 33 in the standard care group.”

TABLE 3 Sensitivity and specificity of RobotReviewer at detecting High RoB

RoB domain	Sensitivity (95% CI)	Specificity (95% CI)
Random sequence generation	0.429 (0.099, 0.816)	0.995 (0.98, 0.999)
Allocation concealment	0.250 (0.0063, 0.806)	0.995 (0.981, 0.999)
Blinding of participants and personnel	0.250 (0.136, 0.396)	0.969 (0.944, 0.985)
Blinding of outcome assessment	0.075 (0.0209, 0.182)	0.991 (0.973, 0.998)
Overall RoB	0.164 (0.1, 0.246)	0.966 (0.936, 0.984)

Abbreviation: RoB, risk of bias.

TABLE 4 Data tables of relevant and irrelevant quotations and RobotReviewer RoB assessment

Random sequence generation				Blinding of participants and personnel			
Quotation	RobotReviewer			Quotation	RobotReviewer		
	Low	Unclear/high	Total		Low	Unclear/high	Total
Irrelevant	5	60	65	Irrelevant	10	260	270
Relevant	147	160	307	Relevant	16	86	102
Total	152	220	372	Total	26	346	372
Allocation concealment				Blinding of outcome assessment			
Quotation	RobotReviewer			Quotation	RobotReviewer		
	Low	Unclear/high	Total		Low	Unclear/high	Total
Irrelevant	16	186	202	Irrelevant	61	201	262
Relevant	89	81	170	Relevant	53	57	110
Total	105	267	372	Total	114	258	372

Abbreviation: RoB, risk of bias.

- *Allocation concealment*: “The velocity was slower, and the maximal amplitude of sway was noticeably lesser among our subjects than in Era’s 7 study.”
- *Blinding of participants and personnel*: “The subjects were instructed to complete as many repetitions as possible until they experienced fatigue.”
- *Blinding of outcome assessment*: “Subjects randomized to the treadmill-walking exercise program were trained as previously described.”

For those studies where irrelevant quotations were selected, RobotReviewer judged 7.7%, 7.9%, 3.7%, and 23.3% of studies as low RoB for random sequence generation, allocation concealment, blinding of participants and personnel, and blinding of outcome assessment, respectively (see Table 4).

4 | DISCUSSION

4.1 | General results

The results of our study indicate that, when compared with human reviewers, RobotReviewer has good agreement for random sequence generation and allocation concealment and fair/poor agreement for blinding participants and personnel, blinding outcome assessment, and overall bias, using two or three categories of RoB decisions. One striking finding was that over 70% of the quotations abstracted by RobotReviewer to support RoB assessment for blinding participants and personnel (72.6%) and blinding outcome assessment (70.4%) were irrelevant. This finding means that none of the RobotReviewer quotations (generally three per domain with expectation that at least one of them would be

the most relevant to support the scoring) was considered relevant for that specific domain in a high percentage of cases.

To our current knowledge, apart from the developers’ studies, only one study¹² has evaluated the performance of RobotReviewer. That study, however, did not analyze the relevance of the supporting quotations. Thus, this present study provides additional, potentially valuable information regarding the ongoing performance assessment of RobotReviewer.

4.2 | Agreement between RobotReviewer and human reviewers

Our study found that the agreement of RoB assessments between RobotReviewer and the consensus of human reviewers varied across domains. Random sequence generation and allocation concealment had the highest agreement; RobotReviewer had less difficulty assessing RoB correctly in these domains. These domains are those most easily rated based on the study reports and those for which the least judgment is required, as documented by previous research.¹³ Thus, we could recommend that RobotReviewer could be useful to evaluate these domains initially and then human reviewers could double-check the software’s score for accuracy.

Agreement between RobotReviewer and the human reviewers for the other domains had only poor/fair agreement. This finding confirms results of previous studies regarding the poor reliability of the Cochrane Collaboration RoB tool in these domains.^{13,18,19,25,26} The evaluation of the blinding of participants and personnel, as well as of the blinding of outcome assessment domains depends on the outcome analyzed. When multiple outcomes are

evaluated in a RCT, RobotReviewer does not indicate which outcome was used for the RoB assessment. Therefore, it is unknown for which outcomes RobotReviewer provides its assessment. In our study, RobotReviewer could not reliably rate RoB assessments, especially for those domains requiring greater interpretation and judgment on the part of human reviewers (blinding of participants and personnel and blinding of outcome assessment).

As previous research demonstrates, variation exists in the level of agreement in RoB assessments between human reviewers.^{18,19} In light of this information, the degree of agreement between RobotReviewer and human reviewers is not surprising. However, since RoB assessments vary and are dependent on the reviewers' training, knowledge, and familiarity with the subject, it remains unclear whether a machine learning tool could replicate that requisite level of expertise and capture these subjective judgments when making RoB assessments.

4.3 | Sensitivity and specificity of RobotReviewer

The sensitivity and specificity of RobotReviewer obtained in our study were similar to the findings presented by Gates et al¹² for random sequence generation and allocation concealment. However, for other domains, the sensitivity and specificity were much lower than previous findings. This disparity could be due to the differing subject areas of the analyzed papers. We selected trials related to physical therapy, and the complexity of these trials could make it more difficult for RobotReviewer to evaluate the risk of bias.

Overall, we found that RobotReviewer had poor sensitivity to detect low RoB across all domains, compared to human reviewers. Similarly, using RobotReviewer as a semi-autonomous tool (to discriminate between high or unclear risk of bias), we were unable to demonstrate an ability to accurately identify high RoB using the output provided by RobotReviewer.

4.4 | Relevance of abstracted quotations

As noted above, a surprising result was the high percentage of irrelevant quotations retrieved by RobotReviewer. Random sequence generation had the smallest proportion of irrelevant quotations (17.5%), making the provided quotes generally useful when using the tool in semi-autonomous manner. However, for many of the other domains, the majority of quotes were irrelevant requiring reviewers to consult the full text version of the manuscript to verify the RoB assessments provided by RobotReviewer, which could make the review process less rather than more efficient.

However, a recent study²⁷ found that using RobotReviewer as a semi-autonomous tool was nonetheless quicker *on average* than no automation by 69 seconds (for all studies analyzed). This 2019 study, outlined in *BMC Medical Informatics and Decision Making*, was performed with a small sample size and used a sample of reviewers not typical of a team of systematic reviewers (5 of the 41 volunteers had no experience with systematic reviews and nine had no experience with the risk of bias tool). In addition, the subject area of the studies was not specified. Although these recent results are encouraging, the study design could be improved to determine how much time is required for a second reviewer to double-check RobotReviewer's evaluations compared to the use of two human reviewers to conduct the process. This comparison is valuable information that future research should consider.

4.5 | Implications for research and practice

RoB assessment is a key process in SRs and is necessary to accurately and effectively synthesize prior research with respect to a health technology or intervention. Slight to poor agreements among RobotReviewer and the human reviewers were obtained for allocation concealment, blinding of participants and personnel, and blinding of outcome assessment. Biases in these domains can have a substantial influence on estimates of treatment effect.^{26,28-31} For example, inadequate allocation concealment or lack of double-blinding can either overestimate treatment effects (by, on average, 18% and 9%, respectively)²⁹⁻³¹ or underestimate treatment effects.^{32,33} At the trial level, over- or underestimates of treatment effects can produce biased or inaccurate results and conclusions in knowledge synthesis products.^{28,29,34-36} These factors can ultimately have repercussions on decision making and quality of patient care, since different assessments could lead to different decisions for clinical practice. Based on these results, we recommend that systematic reviewer teams closely adhere to specific recommendations for using this type of technology; that is, RobotReviewer can help assist with risk of bias assessment of primary studies but cannot replace human evaluations. Thus, reviewers could rely on RobotReviewer's judgment when there is relevant supporting text (eg, "The randomization sequence was generated by computer by the trial statistician"), which might support a "low" risk of bias assessment and the reviewer might safely make a judgment without reading the rest of the article in detail. However, in cases where there is no relevant supporting information, reviewers should consult the full text to ensure the accuracy of RoB assessments.¹⁰ This consultation is in line with the recommendation provided by the developers of the tool.

While RobotReviewer is a novel approach designed to support HTA/SR processes, our experience using the tool highlighted limitations and shortcomings. First, RobotReviewer requires electronic versions of studies, which can be a challenge when working with older studies. Further, while photocopies of older studies can be converted to PDF files, the choice of software to convert these files may have a notable impact on the accuracy and reliability of RobotReviewer (see the Supporting Information regarding the use of PDF or OCR files). Also, RobotReviewer does not differentiate between unclear and high RoB, requiring human reviewers to manually sift through RobotReviewer output to determine a final RoB assessment. Last, we found that the information abstracted by RobotReviewer (including the most relevant quotations) to substantiate RoB assessments was frequently irrelevant. In practice, this outcome would require reviewers to refer back to the original full text, which might decrease overall efficiency. At the current state of the technology, it may be more efficient for researchers to read the full text and complete risk of bias assessments than assess the value of quotations provided by RobotReviewer, when RobotReviewer does not provide relevant text for judging the domain.

4.6 | Artificial intelligence and knowledge synthesis

The ongoing initiatives of applying artificial intelligence to speed up the process of conducting knowledge synthesis are commended. The rapid generation and proliferation of published research studies internationally will increase the need for efficiency conducting evidence reviews in the future. Technological advancements that decrease the time required to complete a HTA/SR¹ have the potential to improve efficiency, making high quality evidence available more rapidly to decision makers. Several technologies have been developed^{4,5,7,8}; however, most of them remain in early stages of testing. More empirical evidence is needed to determine their applicability in practice.

4.7 | Limitations

This study restricted the analysis to a limited number of RCTs in physical therapy and, therefore, the results might not reflect the agreement between RobotReviewer and human reviewers in other areas of research. However, our results are consistent with a previous 2017 study looking at research published across several subject areas.¹² In addition, the version of RobotReviewer available at the time of data collection did not provide RoB assessment of incomplete outcome data and selective

reporting. The developers mentioned that these domains were too cumbersome to analyze using the text from the trials since for selective reporting of outcomes, reviewers should consult a pre-published protocol. Similarly, for incomplete outcome data, reviewers should conduct a numerical evaluation of participant flow through the trial and examine proportions analyzed. In addition to these limitations, we would like to highlight that although weighted kappa is the preferred method to evaluate agreement between different reviewers for ordinal data, kappa is affected by the prevalence of the events (findings) and thus very low values of kappa may not necessarily reflect low rates of agreement.³⁷ This could have happened when analyzing our results for some domains of the risk of bias tool (see online Appendix S2).

4.8 | Future directions

The selection of different outcomes for RoB assessments may have influenced the poor agreement between RobotReviewer and the human reviewers. As noted in the discussion, RobotReviewer did not clearly specify the outcomes used for RoB assessments, and it is uncertain for which outcomes the RoB assessments were applicable. This uncertainty directly reduces reproducibility of RoB assessment for outcome-dependent domains of the tool. In order to improve the accuracy of RobotReviewer, we recommend that the tool be trained to make RoB assessments separately for each outcome analyzed, or at least for the main outcome of the study.

5 | CONCLUSIONS

Agreement between RobotReviewer and human reviewers ranged from poor agreement for blinding of participants and personnel to good agreement for random sequence generation. RobotReviewer selects a high percentage of irrelevant quotations to complement the RoB ratings. This implies that in a large proportion of studies, reviewers will need to read the full-text manuscript to verify whether the assessments provided by RobotReviewer are accurate. Whether this type of confirmation is efficient to speed up the process of quality assessment of RCTs requires further investigation. In addition, RobotReviewer provides assessment of only four domains at this point. Therefore, reviewers must still read the full text of studies to determine the RoB for domains not analyzed by RobotReviewer (incomplete outcome data, selective reporting, and other sources of bias). Thus, it is still limited evidence addressing how much time is required for a second reviewer to double-check RobotReviewer's evaluations compared to the use of two

human reviewers to conduct the process. These multiple aspects, whose evolution promises efficiency, should be tested empirically.

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CONFLICT OF INTEREST

The author reported no conflict of interest.

AUTHOR CONTRIBUTIONS

S.A.-O. and Mr R.C. contributed to study conception and design, data extraction, data analysis and interpretation, and drafting and revision of the manuscript for critical content and approved the final version for publication. S.C. contributed to data extraction, data analysis and interpretation, and revision of manuscript for critical content and approved the final version for publication.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of this article.

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